

Ipopt revisited: Strategies to solve edge cases

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Agenda

- A tour of Ipopt

 - History

 - Techniques applied

- An attempt of a simple IPM algorithm

 - Challenges encountered

 - Results

- A tour of ripopt

- Experiments

 - What can we learn

- Conclusion

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Ipopt: Interior Point OPTimizer

Popular open-source non-linear optimization package

Bindings available for:

- ▶ AMPL (algebraic modeling language)
- ▶ Fortran
- ▶ GAMS (general algebraic modeling system)
- ▶ Julia
- ▶ Matlab / Scilab
- ▶ .NET languages: C#, F# and Visual Basic.NET
- ▶ Python (several wrappers available)
- ▶ R
- ▶ Rust

Ipopt: The origins

- ▶ PhD thesis by Andreas Wächter in 2002 at Carnegie Mellon [1]
- ▶ Since 2002 part of the COIN-OP foundation [2]
- ▶ Original implementation in Fortran 77 (1.x and 2.x), now deprecated
- ▶ Open source since the start
- ▶ Benchmarked against KNITRO (1999) and LOQO (1999) [3, 4]

Ipopt: Subsequent paper

- ▶ Paper by Wächter and Biegler in 2006 [5]
- ▶ More focused on the central algorithm of Ipopt
- ▶ Features described more in detail
 - ▶ Feasibility restoration phase,
 - ▶ Inertia correction,
 - ▶ Various heuristics...
- ▶ More benchmarks: CUTEr test set [6]

Ipopt: C++ version

- ▶ IBM Research decided to invest in an open source re-write of Ipopt in C++
- ▶ With the help of Carl Laird, the code was re-implemented from scratch [7]
- ▶ 3.x published in Aug 2005
- ▶ It is still actively developed on GitHub

Ipopt: Summary of applied techniques

- ▶ Primal-dual interior-point method with filter line-search globalization
- ▶ Sparse symmetric KKT solves with inertia monitoring/correction
- ▶ Inertia correction and regularization to obtain the proper search direction
- ▶ Feasibility restoration phase for difficult or infeasible iterates
- ▶ Second-order correction steps to reduce Maratos-type effects
- ▶ Practical heuristics for robustness and faster performance

Interior Point Methods: Intro

$$\min_{x \in \mathbb{R}^n} \quad \varphi_\mu(x) := f(x) - \mu \sum_{i=1}^n \ln(x^{(i)}) \quad (1)$$

$$\text{s.t.} \quad c(x) = 0 \quad (2)$$

$$x \geq 0 \quad (3)$$

We recover the Karush-Kuhn-Tucker (KKT) conditions for $\mu = 0$:

$$\nabla f(x) + \nabla c(x)\lambda - z = 0 \quad (4)$$

$$c(x) = 0 \quad (5)$$

$$XZe - \mu e = 0 \quad (6)$$

$$x, z \geq 0, \quad \lambda \in \mathbb{R}^m, \quad z \in \mathbb{R}^n,$$

$$Z := \text{diag}(z), e = (1, \dots, 1)^T$$

n : number of primal variables, m : number of equality constraints

KKT system: Derivation

$$F_{\mu}(x, \lambda, z) = \begin{bmatrix} \nabla f(x) + \nabla c(x)\lambda - z \\ c(x) \\ XZe - \mu e \end{bmatrix} = 0$$

Iteratively solve using Newton's method:

- ▶ Fix a starting point (x_k, λ_k, z_k) ,
- ▶ Calculate the deltas for each dimension $(d_k^x, d_k^{\lambda}, d_k^z)$
- ▶ Use the deltas to move to $(x_{k+1}, \lambda_{k+1}, z_{k+1})$ (more later)

$$F_{\mu_j}(x_k, \lambda_k, z_k) + \nabla F_{\mu_j}(x_k, \lambda_k, z_k)(d_k^x, d_k^{\lambda}, d_k^z)^T = 0$$

The term $\nabla F_{\mu}(x, \lambda, z)$ is a 3×3 matrix.

It is the Jacobian of the vector field.

We can move $F_{\mu}(x, \lambda, z)$ to the other side to get the form $Ax = b$.

KKT system: Matrix form

$$\begin{bmatrix} W_k & A_k & -I \\ A_k^T & 0 & 0 \\ Z_k & 0 & X_k \end{bmatrix} \begin{bmatrix} d_k^x \\ d_k^\lambda \\ d_k^z \end{bmatrix} = - \begin{bmatrix} \nabla f(x_k) + A_k \lambda_k - z_k \\ c(x_k) \\ X_k Z_k e - \mu_j e \end{bmatrix}$$

where

$$\begin{aligned} A_k &:= \nabla c(x_k), \\ W_k &:= \nabla_{xx}^2 \mathcal{L}(x_k, \lambda_k, z_k), \\ \mathcal{L}(x, \lambda, z) &:= f(x) + c(x)^T \lambda - z. \end{aligned}$$

But this system can be further simplified!

Multiply out the third equation:

$$\begin{aligned} Z_k d_k^x + X_k d_k^z &= \mu_j e - X_k Z_k e \\ d_k^z &= \mu_j X_k^{-1} e - z_k - X_k^{-1} Z_k d_k^x \\ \Sigma_k &:= X_k^{-1} Z_k \end{aligned}$$

KKT system: Symmetric matrix form

Now d_k^z only appears in the first equation as $-ld_k^z = -d_k^z$ which we move to the other side:

$$\begin{aligned}W_k d_k^x + A_k d_k^\lambda &= -(\nabla f(x_k) + A_k \lambda_k - z_k) + d_k^z \\W_k d_k^x + A_k d_k^\lambda &= -(\nabla f(x_k) + A_k \lambda_k - z_k) + \\&\quad + \mu_j X_k^{-1} e - z_k - \Sigma_k d_k^x \\(W_k + \Sigma_k) d_k^x + A_k d_k^\lambda &= -(\nabla f(x_k) + A_k \lambda_k - \mu_j X_k^{-1} e) \\(W_k + \Sigma_k) d_k^x + A_k d_k^\lambda &= -(\nabla \varphi_{\mu_j}(x_k) + A_k \lambda_k)\end{aligned}$$

Reconstructing the system, we get:

$$\begin{bmatrix} W_k + \Sigma_k & A_k \\ A_k^T & 0 \end{bmatrix} \begin{bmatrix} d_k^x \\ d_k^\lambda \end{bmatrix} = - \begin{bmatrix} \nabla \varphi_{\mu_j}(x_k) + A_k \lambda_k \\ c(x_k) \end{bmatrix} \quad (7)$$

KKT system: Important considerations

$$\begin{bmatrix} W_k + \Sigma_k & A_k \\ A_k^T & 0 \end{bmatrix} \begin{bmatrix} d_k^x \\ d_k^\lambda \end{bmatrix} = - \begin{bmatrix} \nabla \varphi_{\mu_j}(x_k) + A_k \lambda_k \\ c(x_k) \end{bmatrix}$$

- ▶ Symmetric system
- ▶ Matrix in top-left block projected onto the null space of the constraint Jacobian A_k^T is positive definite
- ▶ If A_k does not have full rank, the iteration matrix is singular and a solution may not exist

Solve instead the following system with **inertia correction** for some $\delta_w, \delta_c \geq 0$:

$$\begin{bmatrix} W_k + \Sigma_k + \delta_w I & A_k \\ A_k^T & -\delta_c I \end{bmatrix} \begin{bmatrix} d_k^x \\ d_k^\lambda \end{bmatrix} = - \begin{bmatrix} \nabla \varphi_{\mu_j}(x_k) + A_k \lambda_k \\ c(x_k) \end{bmatrix}$$

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I wanted to see if optimization in pure Rust is possible
memory safety, performance, modern language for low-level applications, ...

I wanted to contribute something back to the community,
i.e, **not** throw away the Python code 1 week after the seminar

The original plan

1. Select the most popular optimization package
2. Implement a simple interior-point method
3. Compare it with Ipopt
4. Profit!



argmin v0.11.0

Numerical optimization in pure Rust

[Homepage](#) [Documentation](#) [Repository](#)

📄 All-Time: 3,049,452

📄 Recent: 531,942

🔄 Updated: 7 months ago

Website [8]

Pure Rust 🦀

We aim at providing a wide range of solvers implemented in pure Rust.

Consistent interface 🐦

Easily plug your optimization problem in different solvers thanks to a consistent interface.

Type agnostic 🔧

Represent your variables, gradients, Jacobians and Hessians using your own types.

Various math backends 🏗️

Use Vecs, ndarray, or nalgebra for behind-the-scenes math. Or implement your own backend.

Observers 👁️

Observe the progress of your optimization by logging to screen or file or implement your own observer.

Checkpointing 🚩

Mitigate the negative effects of crashes in unstable computing environments by saving regular checkpoints.

Extensible 🦋

Most features can be exchanged for user supplied implementations thanks to Rusts powerful generics and traits.

Development framework 🏠

Use argmins tools and ecosystem to implement your own solver – with all the features mentioned above.

discarded several libraries: dealt only with LP (good_lp), used novel algorithms (clarabel), or were too small (interiors)

argmin: Code example

```
1  impl<O, P, G, H, F> Solver<O, IterState<P, G, (), H, (), F>> for Newton<F>
2  where
3      O: Gradient<Param = P, Gradient = G> + Hessian<Param = P, Hessian = H>,
4      P: Clone + ArgminScaledSub<P, F, P>,
5      H: ArgminInv<H> + ArgminDot<G, P>,
6      F: ArgminFloat,
7  {
8      fn next_iter(
9          &mut self,
10         problem: &mut Problem<O>,
11         mut state: IterState<P, G, (), H, (), F>,
12     ) -> Result<(IterState<P, G, (), H, (), F>, Option<KV>), Error> {
13         let param = state.take_param().ok_or_else(argmin_error_closure!(
14             NotInitialized,
15             concat!(
16                 "`Newton` requires an initial parameter vector. ",
17                 "Please provide an initial guess via `Executor`s `configure` method."
18             )
19         ));
20         let grad = problem.gradient(&param)?;
21         let hessian = problem.hessian(&param)?;
22         let new_param = param.scaled_sub(&self.gamma, &hessian.inv()?.dot(&grad));
23         Ok((state.param(new_param), None))
24     }
25 }
```

Math backends

The math backend is implemented in a separate crate
`argmin-math`

It is trait-based: interfaces define the methods required from the
linear algebra backend

- ▶ `nalgebra` [9]
- ▶ `faer` [10]

Add a solver

Add a trait:

```
1  pub trait ArgminLuSolve<T> {
2      /// Solve linear system using LU decomposition.
3      fn lu_solve(&self, rhs: &T) -> Result<T, Error>;
4  }
```

And the corresponding implementation:

```
1  impl<N, D, C, S> ArgminLuSolve<OMatrix<N, D, C>> for SquareMatrix<N, D, S>
2  where
3      N: ComplexField,
4      D: Dim + DimMin<D, Output = D>,
5      C: Dim,
6      S: Storage<N, D, D>,
7      DefaultAllocator: Allocator<N, D, D> + Allocator<N, D, C> + Allocator<N, D>,
8  {
9      #[inline]
10     fn lu_solve(&self, rhs: &OMatrix<N, D, C>) -> Result<OMatrix<N, D, C>, Error> {
11         let lu = LU::new(self.clone_owned());
12         lu.solve(rhs).ok_or_else(|| LuSolveError {}.into())
13     }
14 }
```

No constraints?

```
1  /// Defines equality constraints for a problem.
2  ///
3  /// It follows the form  $h(x) = (0, \dots, 0)$ , where  $h$  is a vector field.
4  pub trait EqualityConstraint {
5      /// Type of the parameter vector
6      type Param;
7      /// Type of the return value of the equality constraints
8      type Output;
9
10     /// Compute all equality constraints
11     fn equality_constraint(&self, param: &Self::Param) -> Result<Self::Output, Error>;
12 }
```

And their derivatives...

```
1  /// Defines the Jacobian of the vector field representing the equality constraints for a  
2  ↳ problem.  
3  pub trait EqualityConstraintJacobian {  
4      /// Type of the parameter vector  
5      type Param;  
6      /// Type of the Jacobian  
7      type Jacobian;  
8      /// Compute the Jacobian of the equality constraints  
9      fn equality_constraint_jacobian(&self, param: &Self::Param) -> Result<Self::Jacobian,  
10         ↳ Error>;  
11  
12     bulk!(equality_constraint_jacobian, Self::Param, Self::Jacobian);  
13 }
```


And the Hessian of the Lagrangian...

```
1  pub trait LagrangianHessian {
2      /// Type of the parameter vector
3      type Param;
4      /// Type of the multipliers associated with equality constraints
5      type MultipliersEq;
6      /// Type of the multipliers associated with inequality constraints
7      type MultipliersIneq;
8      /// Type of the Hessian
9      type Hessian;
10
11     /// Compute the Hessian of the Lagrangian
12     fn lagrangian_hessian(
13         &self,
14         param: &Self::Param,
15         multipliers_eq: &Self::MultipliersEq,
16         multipliers_ineq: &Self::MultipliersIneq,
17     ) -> Result<Self::Hessian, Error>;
18
19     bulk!(
20         lagrangian_hessian,
21         Self::Param,
22         Self::MultipliersEq,
23         Self::MultipliersIneq,
24         Self::Hessian
25     );
26 }
```

And iteration state for NLP...

- ▶ `param: Option<P>`
- ▶ `slacks: Option<S>`
- ▶ `best_param: Option<P>`
- ▶ `best_slacks: Option<S>`
- ▶ `cost: F`
- ▶ `best_cost: F`
- ▶ `target_cost: F`
- ▶ `grad: Option<G>`
- ▶ `lagrangian_hessian: Option<LH>`
- ▶ `equality_constraints: Option<EqC>`
- ▶ `inequality_constraints: Option<IneqC>`
- ▶ `equality_constraint_jacobian: Option<EqCJ>`
- ▶ `inequality_constraint_jacobian: Option<IneqCJ>`
- ▶ `mu: Option<F>`
- ▶ `lambda_eq: Option<LambdaEq>`
- ▶ `lambda_ineq: Option<LambdaIneq>`
- ▶ `alpha_primal: Option<F>`
- ▶ `alpha_dual: Option<F>`
- ▶ `inf_pr: Option<F>`
- ▶ `inf_du: Option<F>`
- ▶ `compl_inf: Option<F>`
- ▶ `iter: u64`
- ▶ `last_best_iter: u64`
- ▶ `max_iters: u64`
- ▶ `counts: HashMap<String, u64>`
- ▶ `counting_enabled: bool`
- ▶ `time: Option<Duration>`
- ▶ `termination_status: TerminationStatus`

Finally! IPM

`nalgebra_backend.rs`: All the linear-algebra-specific logic for `nalgebra`, the only supported backend

`linesearch.rs`: The custom backtracking linear search. The existing implementation could not be reused

`iteration.rs`: A single iteration of the interior-point method

`interiorpointmethod.rs`: The solver. The high-level management of the IPM algorithm.

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riopt: A Rust re-write from February 2026

Coded in ~60/70 hs during the course of 6 weeks heavily relying on Claude Code [11].

Opus 4.6 with a 1M context window. Several agents running in parallel.

License? EPL, same as Ipopt. Author considers it a derivative work.

Linear algebra solver?

- ▶ First relied on a reimplementation of MUMPS [12]
- ▶ Since April 26, it relies on a separate AI-coded solver by the same author [13]

riopt: Licensing

Author's impression:

For small problems there are compelling reasons to be excited about riopt. For larger problems, there is still a lot of work to do to make it competitive.

Interesting insights into thought process documented in the repo.

It is not a "clean room" development.

Caution!

I have not read all the generated code (I don't even really know how to write Rust, but it is readable). It was generated faster than I could read or understand.

riopt: Benchmarks used

Benchmarks and testing are key to make sure that it actually works!

- ▶ Hock-Schittkowski test suite (120 problems) [14, 15] ¹
- ▶ CUTEst 745 problems [16]
- ▶ Electrical grid optimization problems (4 problems) [17] ²
- ▶ Gas pipeline problems (4 problems) [18]
- ▶ Electrolyte thermodynamics problems (13 problems)
- ▶ Parameter estimation for a CHO (Chinese Hamster Ovary) cell kinetic model (1 problem)
- ▶ Water distribution network problems (6 problems)

We will see how much of this holds up to scrutiny...

¹The repo uses a certain smaller set of 120 problems; the papers list 306 problems.

²AC Optimal Power Flow (ACOPF)

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Ipopt's performance can still be improved.

Different methods complement each other: By combining them, edge cases can be solved more efficiently.

Implementation of a state-of-the-art optimization package may be in the reach of modern AI agentic coding tools.

Questions?

Links

Presentation:

https://github.com/hlisdero/JMCS_Seminar_Applied_Optimization

Modified argmin:

<https://github.com/hlisdero/argmin>

riopt:

<https://github.com/jkitchin/riopt>

ipopt-rs:

<https://github.com/elrnv/ipopt-rs>

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